Earth and Planetary Science Letters, 55 (1981) 48–52
Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

Paleomagnetism in the northeastern Central Massif (France): evidence for Carboniferous rotations of the Hercynian orogenic belt

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Received March 16, 1981
Revised version received May 18, 1981

A paleomagnetic study has been carried out on Visean volcanics of northeastern Central Massif. Standard paleomagnetic techniques have been applied to a hundred of samples. Alternating field and thermal demagnetizations yield an unusual direction, $D=78°$, $I=-2°$, $\alpha_{95}=12°$ for the 9 best sites. As the polarity is unknown, two motions can explain this direction, either a clockwise rotation of about 45° or an anticlockwise of 135° relative to the Armorican Massif. In both cases, the paleomagnetic data reveal important deformations of the Hercynian orogenic belt probably due to strike slips in middle Carboniferous times.

1. Introduction

Paleomagnetic studies show that in Devonian times, Laurasia and Gondwana were widely separated by a Hercynian ocean ([1], in [2,3]). This ocean closed during the Carboniferous, but the location, the number and the ages of the suture zones are still speculative. Thus, paleomagnetic studies of formations taken in the middle of the Hercynian orogenic belt would aid in locating these sutures. Inclinations would indicate which continent, Laurasia or Gondwana, the formations belong to; declinations would yield possible deformations, i.e., rotations of the fold belt.

For example, the northern part of the Armorican Massif shows inclinations similar to those of northern Europe and thus has not undergone any large northward drift in Devonian-Carboniferous times [2]. But the same data indicates a slight clockwise rotation of about 15° after the lower Carboniferous.

In northern Spain, declinations follow the Iberian arc [4,5] showing how this arc was formed. The declinations observed in the Westphalo-Stephanian calc-alkaline volcanics from northern

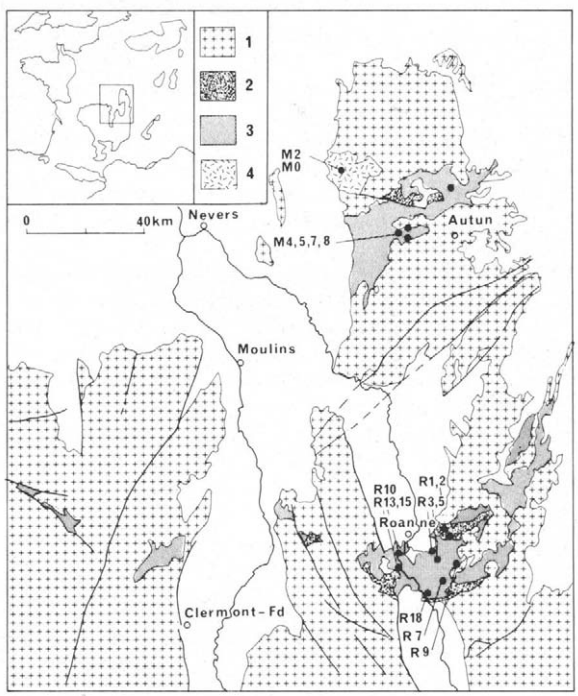


Fig. 1. Geological map of the northeastern Central Massif and sampling sites. *1*=cristalline and metamorphic basement; *2*=Visean sediments; *3*=Visean volcanics; *4*=Late Visean or post-Visean volcanics.

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Corsica to southeastern Sardinia deviate by up to about 90° and yield an anticlockwise rotation during Upper Carboniferous times [6].

All these results reveal the importance of motions that occurred during the Carboniferous in the Hercynian orogenic belt. In order to improve the knowledge of these deformations in other parts of the belt, we have sampled Visean volcanics from the northeastern part of the Central Massif (Fig. 1). These volcanics outcrop also in the Armorican Massif and in the Vosges. Thus, they constitute a good reference line for post-Visean deformations.

2. Geological setting and sampling

The Devono-Dinantian formations of the region of Roanne and the Morvan belong to series of synclinoriums which are generally oriented and folded along SW-NE directions and cut by right-hand and left-hand NNW-SSE strike-slip faults [7].

The Roanne volcanic district shows the following sequence [8,9]:

— Upper Devonian to Lower Visean: slightly metamorphosed andesitic lavas and tuffs (site R1),

— discordance.

No internal deformation effects the following series,

— Middle Visean: basalts interbedded with dated sediments (site R2),

— Middle/Upper Visean: trachy-andesitic latites (R3),

— Upper Visean: a thick series of anthraciferous, dacitic to rhyolitic tuffs (sites R5,7,9,13,15,18) and at the top of the sequence rhyolites (site R10).

In the Morvan, the sequence is similar and begins with basic volcanics in Upper Devonian/Lower Visean times, and continues with anthracitic tuffs (sites M4, 5, 7, 8) and rhyolites during Upper Visean. Some ignimbrites (site M2) younger than the former and lamprophyres of unknown age (M0) have also been sampled.

At each site an average of 6 oriented hand samples were collected. Orientation was done by magnetic and sun compasses. The tectonic setting of the volcanic formations was difficult to estimate and it was often impossible to see bedding planes. Thus we had to rely on "flammes" and mineral layering. For several sites, we overcame the lack of field data by anisotropy of susceptibility measurements. We observed that in such volcanic formations the least susceptibility axis is usually perpendicular to the bedding plane. For a few sites we used this as a determination of the bedding plane.

3. Measurements and demagnetizations

The intensity of NRM is generally low (10^{-2} to 10^{-3} A/m). Several sites were discarded because of the low intensity of their magnetization (less than 10^{-3} A/m). apparently due to weathering effects. The susceptibility ranges from 5×10^{-5} to 25×10^{-5} SI units. 124 samples have been further

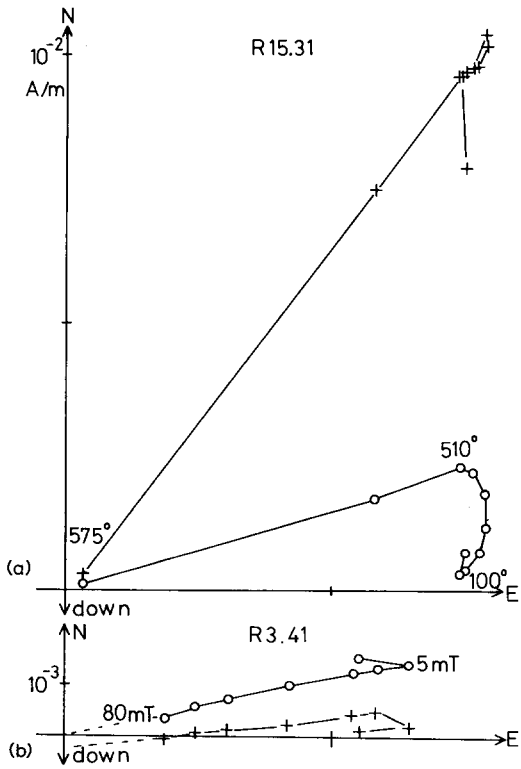


Fig. 2. Typical demagnetizations: high thermal (specimen R15.31) and alternating field (specimen R3.41) processes; dots in the horizontal plane, crosses in the E-W-down plane.

TABLE 1

Directions of characteristic magnetizations

Site	Rock type	Age	A	N/N_0	n	D (°)	I (°)	Dip	D_c (°)	I_c (°)	k	α_{95} (°)
R1	andesites	Upper Devonian Lower Visean	1?	2/3	3	77.5	−16	?			46	18
R2	basalts	Middle Visean	1?	3/5	7	61	−13	?			43	9
R3	latites	Middle/Upper Visean	1	6/6	7	77	−9	10°N345°	79	−8	238	4
R7	anthr. tuffs	Upper Visean	1	2/6	4	87	−13	35°N 0°	95	−12	27	18
			1	4/6	6	93	−17	35°N0°	101	−12	11	21
R9	anthr. tuffs	Upper Visean	1	6/6	9	37	78	70°N60°	60	9	64	6.5
R5	anthr.tuffs	Upper Visean	1	6/6	7	86	−72	55°N240°	70	−19	260	4
R13	anthr. tuffs	Upper Visean	1	7/7	7	205	78	80°N245°	238	1	170	5
R15	anthr. tuffs	Upper Visan	2	3/7	5	46	−43	70°N310°	82	−9	41	12
			1	3/7	6	79	−48	70°N310°	98	8	69	8
R18	anthr. tuffs	Upper Visean	2	5/5	7	36	−26	70°N305°	61	−9	26	13
			1	5.5	8	55	−26	70*N305°	67	7	79	6
M4	anthr. tuffs	Upper Visean	1	5.6	7	97	31	50°N315°	50	58	72	7
M5	anthr. tuffs	Upper Visean	1	6.7	6	90	52	35°N330°	37	53	215	5
M7	anthr. tuffs	Upper Visean	1	6/6	8	265	12	20°N0°?	269	13	18	14
M8	anthr. tuffs	Upper Visean	1	5/6	5	268	1	40°N330°?	263	−15	89	8
R10	rhyolite	Upper Visean	1	5/6	5	338	60	75*N70?	44	13	42	12
M2	ignimbrite	Upper Visean or later	1	5/6	6	208	−29	40°N270°	180	−40	125	6
M0	lamprophyre	later than M2	1	6/6	8	187	−19	30°N192°	179	−10	59	10
Mean (R3, R7, R9, R5, R13, R15, R18, M7, M8) without tilt corrections					9	76	−25				3	33
Mean (idem), with tilt corrections					9				78	−2	19	12

A = primary (1) or secondary (2) magnetization; N = number of samples used; N_0 = total number of samples collected; n = number of specimens used for statistic computation; D, I = declination and inclination after demagnetization; Dip = inclination and direction of maximal slope; D_c, I_c = declination and inclination after bedding correction; k, α_{95} = Fisher's statistics parameters.

analysed, and from each, one or two specimens were demagnetized by alternating fields of up to 150 mT (Fig. 2b). At least one specimen per site was thermally demagnetized (Fig. 2a). Anthraciferous tuffs display magnetite blocking temperatures (580°) while some ignimbrites (M2) and the lamprophyres (M0) exhibit haematite temperatures (670°). After removal of the characteristic magnetization carried by magnetite, there is often a small spurious residual magnetization with blocking temperatures typical of titano-haematites and with diverging directions. The same phenomenon was observed on similar volcanics of Sardinia [6,10]. This residual magnetization is attributed to low-temperature oxidation of titanomagnetites by subsequent weathering. The main magnetization directions which are carried by magnetite are usually well grouped and the angle of 95% confidence α_{95}) remains usually lower than 10° (Table 1). The main difficulty comes from the tectonic corrections. The dip is often high, sometimes nearly vertical. All corrections were done assuming that the fold axes were horizontal. This may explain the scatter of the mean sites directions, particularly with declinations (Table 1) and consequently of the poles (Fig. 3).

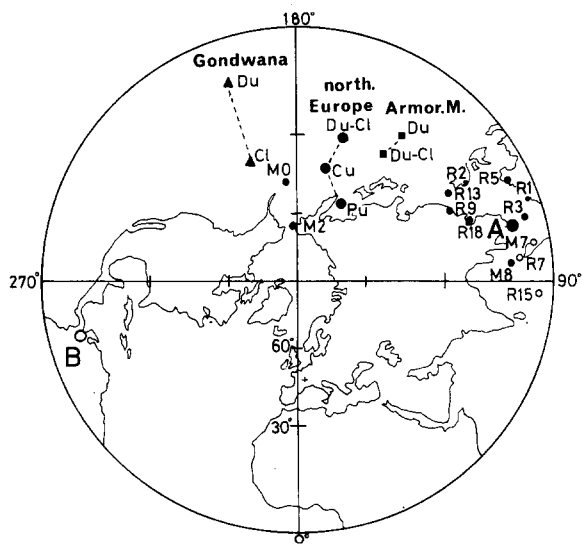


Fig. 3. Geomagnetic pole positions of the northeastern Central Massif (*A* in case of normal polarity, *B* in case of reversed), the Armorican Massif [2], northern Europe [1] and Gondwana [1].

4. Discussion and interpretation

Two sites from Morvan (M4 and M5) show high inclinations difficult to explain in spite of the good quality of the demagnetizations. Possible explanations may be: an incorrect tilt correction, a reheating due to subsequent emplacements of microgranites, or a spurious field direction due to a cooling during a reversal. The younger formations from Morvan (sites M2 and M0) have magnetization directions and poles quite near to those from stable Europe during Late Carboniferous/Early Permian (Table 1, Fig. 3).

For the Visean series of the Roannais (sites R1, 2, 3, 5, 7, 9, 13, 15, 18) and Morvan (sites M7, 8) the fold test is positive (Table 1). If we exclude sites M4 and M5 by reason of their high inclinations and sites R1, R2 and R10 for which no tilt correction was possible, the mean direction for 9 sites is $D = 178°$, $I = -2°$, $\alpha_{95} = 12°$. The low inclinations and the unusual declinations lead to an undetermined polarity of the mean direction. A normal direction (pole A) implies that Morvan and Roannais would have undergone a clockwise rotation of about 45° relative to the Armorican Massif and 60° relative to northern Europe. On the other hand, a reversed direction (pole B) suggests that the rotation would be much more important and anticlockwise. In both cases, the area would have been situated near the paleo-equator. In spite of this indeterminancy, one notices that since Upper Visean times, the northern Central Massif has not undergone any significant northward drift relative to northern Europe. The rotation may be a result of Late Hercynian strike slips, but this has to be supported by further investigations on neighbouring series of Central Massif, southern Vosges and eastern Armorican Massif where similar volcanic series outcrop.

Acknowledgements

We are indebted to Dr. R. Coulon of the University of Reims, who guided us in the Morvan, and Dr. R.F. Rubiello and Dr. J. Bertaux who gave us useful indications of the volcanics of Roanne. Dr. R. Montigny kindly read the manuscript.

This work was supported by E.R.A. 683 (now part of LA 323) from C.N.R.S. and A.T.P. "Geodynamique" from I.N.A.G.

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